Chapter 8. Generating Functions

A Function Is Worth Many Numbers

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A generating function stores a whole sequence; operations on it correspond to combinatorial constructions.

8.1 Ordinary generating functions

8.1.1 Recurrence relations and generating functions

Motivation. Summing a recurrence times x^{n+1} packs it into one equation for a power series.

Frogs quadruple yearly and 100 are removed each year: $a_0 = 50$ and

$$a_{n+1} = 4a_n - 100 \qquad (n \geq 0). \tag{8.1}$$

Definition 8.1 (Ordinary generating function)

For a sequence $(f_n)_{n\geq 0}$ of reals, the **formal power series** $F(x) = \sum_{n\geq 0} f_n x^n$ is its **ordinary generating function**.

Formal power series need not converge (e.g. $\sum n!x^n$); equality means equal coefficients.

Multiplying (8.1) by x^{n+1} and summing, with $G(x) = \sum a_n x^n$:

$$G(x) - a_0 = 4xG(x) - \frac{100x}{1-x}, \qquad G(x) = \frac{50}{1-4x} - \frac{100x}{(1-x)(1-4x)}. \tag{8.2}$$

Partial fractions give $\frac{100x}{(1-x)(1-4x)} = \frac{100}{3}\left(\frac{1}{1-4x} - \frac{1}{1-x}\right)$. Since $\frac{1}{1-4x} = \sum 4^n x^n$,

$$a_n = 50 \cdot 4^n - 100 \cdot \frac{4^n - 1}{3}, \tag{8.3}$$

which indeed satisfies $a_0 = 50$ and (8.1).

The method. (1) $G(x) = \sum a_n x^n$. (2) Multiply the recurrence by x^{n+k}, sum over $n \geq 0$. (3) Solve for G. (4) Read off coefficients.

Example 8.2

\$1000 earns 5% interest per year and \$500 is deposited each year: $a_0 = 1000$, $a_{n+1} = 1.05a_n + 500$.

Solution. $G(x) - 1000 = 1.05xG(x) + \frac{500x}{1-x}$, so

$$G(x) = \frac{1000}{1-1.05x} + \frac{500x}{(1-x)(1-1.05x)} = \frac{1000}{1-1.05x} + \frac{10000}{1-1.05x} - \frac{10000}{1-x},$$

and $a_n = 11000 \cdot 1.05^n - 10000$.

Example 8.3

$a_0 = 0$, $a_1 = 1$, $a_{n+2} = 3a_{n+1} - 2a_n$. Multiplying by x^{n+2} and summing: $G - x = 3xG - 2x^2G$. So

$$G(x) = \frac{x}{1-3x+2x^2} = \frac{1}{1-2x} - \frac{1}{1-x}, \qquad a_n = 2^n - 1.$$

8.1.2 Products of generating functions

Motivation. Products convolve coefficients: cut $[n]$ into two intervals with independent structures.

Lemma 8.4

If $A(x) = \sum a_n x^n$, $B(x) = \sum b_n x^n$ and $c_n = \sum_{i=0}^{n} a_i b_{n-i}$, then $A(x)B(x) = C(x) := \sum c_n x^n$.

Proof. $a_i x^i \cdot b_j x^j$ contributes to x^n iff $j = n - i$. ■

Theorem 8.5 (Product formula)

Let a_n (b_n) be the number of structures of a first (second) kind on an n-element set. Let c_n be the number of ways to cut $[n]$ into the intervals $S = \{1, \dots, i\}$ and $T = \{i+1, \dots, n\}$ (possibly empty) and build a structure of the first kind on S and of the second kind on T. Then $A(x)B(x) = C(x)$.

Proof. Cut point i gives $a_i b_{n-i}$ choices; apply Lemma 8.4. ■

By induction, likewise for any fixed number of intervals.

Example 8.6

An n-day semester is split into a first part (k days, one holiday) and a second ($n-k$ days, two holidays). The number of plans is $f_n = \binom{n+1}{4}$.

Solution. $f_n = \sum_k k\binom{n-k}{2}$, with $A(x) = \sum_{k\geq 1} kx^k = \frac{x}{(1-x)^2}$ and $B(x) = \sum_{m\geq 2}\binom{m}{2}x^m = \frac{x^2}{(1-x)^3}$:

$$F(x) = A(x)B(x) = \frac{x^3}{(1-x)^5} = x^3 \sum_{n\geq 0} \binom{n+4}{4} x^n = \sum_{n\geq 3} \binom{n+1}{4} x^n.$$

using $(1-x)^{-5} = \sum_n \binom{n+4}{4}x^n$ (Theorem 4.15 or 3.21).

Example 8.7

With any set of days chosen in each part, each part has $C(x) = \sum 2^k x^k = \frac{1}{1-2x}$, so $F(x) = \frac{1}{(1-2x)^2} = \frac{1}{2}C'(x) = \sum_n (n+1)2^n x^n$, and $g_n = (n+1)2^n$.

Example 8.8

Three parts with any, an odd, and an even number of holidays. An m-set ($m \geq 1$) has 2^{m-1} odd and 2^{m-1} even subsets (Theorem 4.2), giving $\frac{1}{1-2x}$, $\frac{x}{1-2x}$ and $1 + \frac{x}{1-2x} = \frac{1-x}{1-2x}$, and

$$G(x) = \frac{x(1-x)}{(1-2x)^3} = -\frac{1}{4}\cdot\frac{1}{1-2x} + \frac{1}{4}\cdot\frac{1}{(1-2x)^3}.$$

With $(1-2x)^{-3} = \sum_n \binom{n+2}{2}2^n x^n$ this gives $g_n = \frac{1}{4}(\binom{n+2}{2} - 1)2^n = 2^{n-3}n(n+3)$.

Example 8.9

With $p_{\leq k}(n)$ the number of partitions of n into parts of size at most k,

$$\sum_{n\geq 0} p_{\leq k}(n)x^n = \prod_{i=1}^{k} \frac{1}{1-x^i} = (1+x+x^2+\cdots)(1+x^2+x^4+\cdots)\cdots(1+x^k+x^{2k}+\cdots). \quad (8.4)$$

Solution. Picking x^{ij_i} from factor i with $\sum_i ij_i = n$ corresponds bijectively to the partition with j_i parts i. By Theorem 5.17 this also counts partitions into at most k parts.

Example 8.10

For all partitions,

$$\sum_{n\geq 0} p(n)x^n = \prod_{k\geq 1} \frac{1}{1-x^k}, \tag{8.5}$$

well defined since only the first n factors affect x^n.

Example 8.11 (Euler)

The number $p_{\text{odd}}(n)$ of partitions of n into odd parts equals the number $p_d(n)$ of partitions into distinct parts.
Solution.

$$\sum_n p_{\text{odd}}(n)x^n = \prod_{i \text{ odd}} \frac{1}{1-x^i}, \qquad \sum_n p_d(n)x^n = \prod_{i\geq 1}(1+x^i) = \prod_{i\geq 1}\frac{1-x^{2i}}{1-x^i}.$$

The numerators cancel the even denominators, leaving $\prod_{i \text{ odd}} \frac{1}{1-x^i}$.

8.1.2.1 The Catalan numbers

Let c_n ($c_0 = 1$) count **good sequences** $b_1, \dots, b_{2n} \in \{\pm 1\}$ with $\sum_j b_j = 0$ and all partial sums ≥ 0 (coins into and out of a jar, never negative).

Decomposition. For $n > 0$ let $2i$ be the first return to 0. Days $2i+1, \dots, 2n$ form a good sequence; days $1, \dots, 2i$ form a **very good** one (partial sums positive inside), which minus its first $+1$ and last -1 is a good sequence of length $2(i-1)$. So very good sequences have generating function $xC(x)$, and by the product formula

$$C(x) - 1 = xC(x) \cdot C(x), \qquad \text{i.e.} \quad xC(x)^2 - C(x) + 1 = 0. \tag{8.6}$$

Of the roots $\frac{1\pm\sqrt{1-4x}}{2x}$ only the minus sign is a power series (the plus sign has a pole at 0):

$$C(x) = \frac{1-\sqrt{1-4x}}{2x}. \tag{8.7}$$

By Theorem 4.15, $\sqrt{1-4x} = \sum_n \binom{1/2}{n}(-4x)^n = 1 - 2\sum_{n\geq 1} \frac{1}{n}\binom{2n-2}{n-1}x^n$ (as in Example 4.16), so

$$C(x) = \sum_{n\geq 0} \frac{1}{n+1}\binom{2n}{n}x^n, \qquad c_n = \frac{1}{n+1}\binom{2n}{n}.$$

the **Catalan numbers** $1, 1, 2, 5, 14, 42, \dots$ (Chapter 14).



Figure. *A good sequence as a path above the axis; cutting at the first return gives $C - 1 = xC \cdot C$.*

8.1.3 Compositions of generating functions

Motivation. Cutting into any number of nonempty intervals corresponds to substitution of series.

$F(G(x)) = \sum f_n G^n$ is well defined when G has constant term 0, since then $x^m \mid G^m$.

Definition 8.12 (Composition)

For a formal power series $F(x) = \sum f_n x^n$ and G with constant term 0, $F(G(x)) := \sum_{n\geq 0} f_n G(x)^n$.

Theorem 8.13

Let a_n count structures of some kind on an n-set, with $a_0 = 0$. Let h_n ($h_0 = 1$) count the ways to cut $[n]$ into any number of nonempty intervals and put such a structure on each. Then $H(x) = \frac{1}{1-A(x)}$.

Proof. $A(x)^k$ counts cuts into k intervals with structures (product formula; nonempty since $a_0 = 0$), and $x^k \mid A^k$, so $H = 1 + \sum_{k\geq 1} A^k = \frac{1}{1-A}$. ■

(With empty intervals there would be infinitely many cuts.)

Example 8.14

n soldiers in a line are split into nonempty units and a commander is named in each. The number of ways is $h_n = \frac{1}{\sqrt{5}}(\alpha^n - \beta^n)$ for $n \geq 1$, with $\alpha, \beta = \frac{3\pm\sqrt{5}}{2}$.
Solution. $a_k = k$, $A(x) = \frac{x}{(1-x)^2}$, $H(x) = 1 + \frac{x}{1-3x+x^2}$; with $\alpha\beta = 1$ the roots of $x^2 - 3x + 1$,

$$\frac{1}{1-3x+x^2} = \frac{1}{\sqrt{5}}\left(\frac{\alpha}{1-\alpha x} - \frac{\beta}{1-\beta x}\right) = \frac{1}{\sqrt{5}}\sum_n (\alpha^{n+1} - \beta^{n+1})x^n.$$

So $h_n = \frac{1}{\sqrt{5}}(\alpha^n - \beta^n)$: $1, 3, 8, 21, 55, \ldots$.

Theorem 8.15 (Compositional formula)

Let a_n count structures of a first kind on an n-set ($a_0 = 0$) and b_n structures of a second kind ($b_0 = 1$). Let g_n ($g_0 = 1$) count the ways to cut $[n]$ into nonempty intervals, put a first-kind structure on each interval and a second-kind structure on the set of intervals. Then $G(x) = B(A(x))$.

Proof. k intervals contribute $b_k A(x)^k$ (product formula), so $G = \sum_k b_k A(x)^k = B(A(x))$. ■

Example 8.16

Split the line into units and choose any subset of units for night duty: $a_k = 1$ ($k \geq 1$), $b_m = 2^m$, $A = \frac{x}{1-x}$, $B = \frac{1}{1-2x}$,

$$G(x) = \frac{1}{1-\frac{2x}{1-x}} = \frac{1-x}{1-3x} = 1 + \sum_{n\geq 1} 2\cdot 3^{n-1}x^n.$$

8.2 Exponential generating functions

8.2.1 Recurrence relations and exponential generating functions

Motivation. Dividing by $n!$ tames fast-growing sequences and absorbs factors $(n+1)$.

Example 8.17

$a_0 = 1$, $a_{n+1} = (n+1)(a_n - n + 1)$ for $n \geq 0$. Then $a_n = n! + n$.

Definition 8.18 (Exponential generating function)

For a sequence $(f_n)_{n\geq 0}$, the formal power series $F(x) = \sum_{n\geq 0} f_n \frac{x^n}{n!}$ is its **exponential generating function** (the constant sequence 1 has e^x).

Solution of Example 8.17. Multiply by $\frac{x^{n+1}}{(n+1)!}$ and sum:

$$\sum_{n\geq 0} a_{n+1}\frac{x^{n+1}}{(n+1)!} = \sum_{n\geq 0} a_n \frac{x^{n+1}}{n!} - \sum_{n\geq 0}(n-1)\frac{x^{n+1}}{n!}, \tag{8.8}$$

i.e. $A(x)(1-x) = 1 + xe^x(1-x)$, so

$$A(x) = \frac{1}{1-x} + xe^x = \sum_{n\geq 0} n!\frac{x^n}{n!} + \sum_{n\geq 1} n\frac{x^n}{n!}.$$

Example 8.19

$f_0 = 0$, $f_{n+1} = 2(n+1)f_n + (n+1)!$. Multiplying by $\frac{x^{n+1}}{(n+1)!}$ and summing: $F(x) = 2xF(x) + \frac{x}{1-x}$. So $F(x) = \frac{x}{(1-x)(1-2x)} = \sum_n (2^n - 1)x^n$, and $f_n = (2^n - 1)\, n!$.

8.2.2 Products of exponential generating functions

Motivation. Exponential products split $[n]$ into two *arbitrary* subsets.

Lemma 8.20

For exponential generating functions $A(x) = \sum a_i \frac{x^i}{i!}$, $B(x) = \sum b_k \frac{x^k}{k!}$ and $c_n = \sum_{i=0}^{n} \binom{n}{i} a_i b_{n-i}$, we have $A(x)B(x) = C(x) := \sum c_n \frac{x^n}{n!}$.

Proof. $a_i \frac{x^i}{i!} \cdot b_j \frac{x^j}{j!} = a_i b_j \binom{i+j}{i} \frac{x^{i+j}}{(i+j)!}$. ∎

Theorem 8.21 (Product formula, exponential version)

Let a_n, b_n count structures of two kinds on n-sets, and let c_n count the ways to split $[n]$ into disjoint subsets $S \cup T = [n]$ (no other restriction) and build a first-kind structure on S and a second-kind structure on T. Then $A(x)B(x) = C(x)$ for the exponential generating functions.

Proof. $|S| = i$: $\binom{n}{i} a_i b_{n-i}$ choices; apply Lemma 8.20. ∎

Example 8.22

Split n players into two lined-up groups, the first wearing one of three shirt colours: $a_k = k!\, 3^k$, $A = \frac{1}{1-3x}$; $b_m = m!$, $B = \frac{1}{1-x}$;

$$C(x) = \frac{1}{(1-3x)(1-x)} = \frac{3}{2}\cdot\frac{1}{1-3x} - \frac{1}{2}\cdot\frac{1}{1-x} = \sum_n \frac{3^{n+1}-1}{2}x^n,$$

and $c_n = n!\,\frac{3^{n+1}-1}{2}$.

Example 8.23

For fixed k, the exponential generating function of the Stirling numbers of the second kind is

$$S_k(x) := \sum_{n\geq k} S(n,k)\frac{x^n}{n!} = \frac{1}{k!}(e^x-1)^k. \tag{8.9}$$

Solution. A nonempty block has generating function e^x-1; $(e^x-1)^k$ counts k *ordered* blocks, each partition $k!$ times. Expanding $(e^x-1)^k = \sum_i \binom{k}{i}(-1)^i e^{(k-i)x}$ recovers Theorem 7.5.

Differentiation shifts: $\left(\sum a_n \frac{x^n}{n!}\right)' = \sum a_{n+1}\frac{x^n}{n!}$.

Example 8.24

The exponential generating function of the Bell numbers is $B(x) = e^{e^x-1}$.
Solution. Summing Theorem 5.12 times $\frac{x^n}{n!}$: $B'(x) = B(x)e^x$ (Lemma 8.20), so $\ln B = e^x + C$, and $B(0) = 1$ gives $C = -1$.

8.2.3 Compositions of exponential generating functions

Motivation. Partitioning into structured blocks exponentiates the generating function.

Theorem 8.25 (Exponential formula)

Let a_n count structures on n-sets ($a_0 = 0$) and let h_n ($h_0 = 1$) count the ways to partition $[n]$ into any number of nonempty blocks and put a structure on each block. Then $H(x) = e^{A(x)}$ for the exponential generating functions.

Proof. $A(x)^k$ counts k *ordered* structured blocks (Theorem 8.21), so $A^k/k!$ counts unordered ones; $x^k \mid A^k$, so $H = 1 + \sum_{k\geq 1}\frac{A^k}{k!} = e^{A(x)}$. ∎

Integrating $\frac{1}{1-x} = \sum x^n$ termwise gives

$$\sum_{k\geq 1}\frac{x^k}{k} = \ln\frac{1}{1-x}.$$

Example 8.26

Groups at circular tables: $a_k = (k-1)!$ (Example 6.10), $A = \ln\frac{1}{1-x}$, $H = \frac{1}{1-x}$, $h_n = n!$ (a permutation is a set of cycles).

Example 8.27

Partitions into blocks of sizes 3, 4, 9 have exponential generating function $e^{x^3/3!+x^4/4!+x^9/9!}$ (exponential formula with $A = x^3/3!$ etc., combined by the product formula).

Theorem 8.28 (Compositional formula, exponential version)

Let a_n count first-kind structures ($a_0 = 0$) and b_n second-kind structures ($b_0 = 1$). Let g_n count the ways to partition $[n]$ into nonempty blocks, put a first-kind structure on each block and a second-kind structure on the set of blocks. Then $G(x) = B(A(x))$.

Proof. k blocks contribute $b_k \frac{A(x)^k}{k!}$ (Theorem 8.21), so $G = B(A(x))$, defined since $a_0 = 0$. ■

Example 8.29

n cards split into ordered subsets of even size, the subsets in a line: $a_n = n!$ for even $n \geq 2$ (else 0), $A = \frac{x^2}{1-x^2}$; $b_n = n!$, $B = \frac{1}{1-x}$;

$$G(x) = \frac{1}{1 - \frac{x^2}{1-x^2}} = \frac{1-x^2}{1-2x^2} = 1 + \sum_{m \geq 0} 2^m x^{2m+2},$$

so $g_n = 2^{n/2-1}\, n!$ for even $n \geq 2$, 0 for odd n.